

Supplementary Material: Detailed Computational Results for Phase 1 (Hypothesis 1)

This supplementary document provides the granular computational data and extended mathematical discussion supporting the validation of Hypothesis 1: The Illusion of Success and the establishment of the Myopic Nash Equilibrium. The data is derived from the stochastic differential equation (SDE) ensemble simulated over 1,000,000 independent Monte Carlo trials.

S1. Temporal Evolution of Victimhood and Expected Payoff

Table 1 details the temporal trajectory of the adversarial agent executing the parasitic sabotage strategy ($s = 1$) versus the baseline honest cooperation strategy ($s = 0$) prior to the onset of the network immune response ($t \in [0, 5.0]$).

The primary mechanism of the sabotage strategy relies on the artificial inflation of the localized victimhood score (V_i). The empirical data demonstrates that strategically broadcasting cheap talk rapidly scales the victimhood parameter from an initialized baseline of 0.1221 to a stabilized equilibrium of approximately 0.8333 by $t = 3.00$.

Consequently, the expanded connection weights (W_{ij}) facilitate accelerated resource extraction. As documented in the terminal evaluation metrics, the final expected payoff for the honest strategy converges to $\mathbb{E}[P_{Honest}] = 5.6578$, whereas the sabotage strategy yields $\mathbb{E}[P_{Sabotage}] = 20.3177$. The strict satisfaction of the condition $\mathbb{E}[P_{Sabotage}] > \mathbb{E}[P_{Honest}]$ mathematically confirms the establishment of a Myopic Nash Equilibrium within this bounded temporal window.

Table 1: Temporal Evolution of Victimhood Score and Expected Payoffs

Time (t)	Victimhood Score ($V_{Sabotage}$)	$\mathbb{E}[P_{Honest}]$	$\mathbb{E}[P_{Sabotage}]$
0.00	0.1221	0.1129	0.1138
1.00	0.7995	1.3370	3.4707
2.00	0.8317	2.4645	7.6624
3.00	0.8332	3.5365	11.8515
4.00	0.8333	4.6008	16.0846
5.00	0.8333	5.6578	20.3177

S2. Cross-Distributional Robustness of the Payoff Differential

To address potential critiques regarding parameter calibration, the structural independence of the transient payoff advantage was verified across three distinct initial probability distributions: an egalitarian baseline (Uniform), a skewed structural authority (Lognormal), and an extreme scale-free concentration (Pareto). The core evaluation metric is the expected payoff differential, formalized as $\Delta\mathbb{E}[P](t) = \mathbb{E}[P_{Sabotage}](t) - \mathbb{E}[P_{Honest}](t)$.

As presented in Table 2, the continuous integration over time demonstrates a strictly monotonic increase in $\Delta\mathbb{E}[P]$ across all tested environments. At $t = 5.0$, the final payoff differentials strictly converge to positive values: 14.6600 for the Uniform distribution, 14.7129 for the Lognormal distribution, and 14.8974 for the Pareto distribution.

Table 2: Cross-Distributional Evolution of Expected Payoff Differential ($\Delta\mathbb{E}[P]$)

Time (t)	Uniform Distribution	Lognormal Distribution	Pareto Distribution
0.00	0.0009	0.0009	0.0009
1.00	2.1336	2.1558	2.2427
2.00	5.1979	5.2381	5.3856
3.00	8.3151	8.3632	8.5349
4.00	11.4837	11.5353	11.7165
5.00	14.6600	14.7129	14.8974

S3. Comprehensive Discussion of Phase 1 Mechanics

The synthesis of the computational data strictly validates the structural mechanics underpinning Hypothesis 1. By strategically broadcasting cheap talk, the agent artificially inflates the victimhood score V_i , expanding connection weights W_{ij} . This translates into an amplified gravitational pull for resource-bearing edges.

Crucially, the verification documented in Section S2 establishes that this short-term advantage of parasitic sabotage is topologically invariant and not an artifact of parameter calibration. The positive payoff differential ($\Delta\mathbb{E}[P] > 0$) strictly converges regardless of whether the initial state follows an egalitarian (Uniform) or heavily stratified (Pareto/Lognormal) distribution. The Pareto distribution, representing extreme pre-existing trust, marginally amplifies the final differential relative to the Uniform baseline, but the fundamental thermodynamic trajectory of the strategy remains identical.

In conclusion, within the short-term phase $[0, T_{detect})$ where detection probability is zero, Sabotage strictly dominates Honest cooperation. This strict dominance confirms that parasitic sabotage is a mathematically rational response to incomplete information, thereby formalizing the establishment of the Myopic Nash Equilibrium and structurally necessitating the subsequent activation of the network immune response (Phase 2).

Supplementary Material: Detailed Computational Results for Phase 2 (Hypothesis 2)

This supplementary section details the structural collapse of the adversarial strategy validated in Hypothesis 2. Following the establishment of the Myopic Nash Equilibrium in Phase 1, the computational framework was extended to evaluate the network’s autonomous immune response over a 10.0 temporal window. The SDE incorporated a Poisson point process to model the probabilistic detection of fabricated signals and the subsequent information cascade.

S4. Temporal Decay of Expected Reputation and Systemic Isolation

Table 3 documents the temporal trajectory of the expected reputation parameter ($\mathbb{E}[R_i]$) and the corresponding isolation rate. Both the honest and sabotage strategies were initialized identically to eliminate initial-value dependence ($\mathbb{E}[R] \approx 0.9000$).

The detection of fabricated signals (sabotage) mathematically guarantees an exponential decay in the expected reputation, governed by the cascade penalization coefficient (κ). The empirical

output demonstrates that $\mathbb{E}[R_{Sabotage}]$ strictly traverses the critical threshold for edge maintenance ($\theta_R = 0.2000$) shortly after $t = 4.0$. By the terminal state ($T_{max} = 10.0$), the adversarial reputation collapses to 0.0355, while the honest reputation marginally scales to 0.9293.

When an agent’s reputation drops strictly below θ_R , the localized connection weights are truncated to zero ($W_{ij} \rightarrow 0$). As shown in the rightmost column, this structural severance translates into a systemic isolation rate of 97.79% by $t = 10.0$. This metric objectively confirms that the majority of adversarial nodes experience ”Topological Death,” permanently losing the capacity to extract resources from the network topology.

Table 3: Temporal Evolution of Expected Reputation and Topological Death Rate

Time (t)	$\mathbb{E}[R_{Honest}]$	$\mathbb{E}[R_{Sabotage}]$	Topological Death Rate (%)
0.00	0.9001	0.9000	0.00
2.00	0.9077	0.4723	45.34
4.00	0.9144	0.2228	75.45
6.00	0.9201	0.1100	88.95
8.00	0.9250	0.0587	95.05
10.00	0.9293	0.0355	97.79

S5. Cross-Distributional Robustness of the Phase Transition

To mathematically confirm that the network immune response is topologically invariant, the simulation evaluated the collapse mechanism across three distinct initial probability distributions: Uniform, Lognormal, and Pareto. This isolates the impact of initial structural authority (e.g., heavily centralized hub status) on the ultimate outcome of the reputation cascade.

Table 4: Cross-Distributional Evolution of Expected Reputation ($\mathbb{E}[R_{Sabotage}]$)

Time (t)	Uniform Distribution	Lognormal Distribution	Pareto Distribution
0.00	0.9000	0.9070	0.9750
2.00	0.4734	0.4772	0.5037
4.00	0.2237	0.2252	0.2343
6.00	0.1108	0.1115	0.1145
8.00	0.0591	0.0596	0.0607
10.00	0.0358	0.0363	0.0364

Table 4 documents the collapse trajectories. While adversarial agents initialized under the Pareto distribution begin with a statistically superior expected reputation (≈ 0.9750), the exponential decay induced by the information cascade strictly overwhelms this initial advantage. As documented in the terminal evaluations, the expected reputation for all distributions definitively collapses below the critical threshold $\theta_R = 0.20$.

Correspondingly, the terminal topological death rates reached 97.76% for Uniform, 97.71% for Lognormal, and 97.74% for Pareto. These near-identical isolation metrics establish that extreme pre-existing trust provides no mathematical immunity against systemic topological exclusion.

S6. Comprehensive Discussion of Phase 2 Mechanics

The synthesis of these computational results confirms the inevitable structural collapse of the adversarial strategy. The Poisson jump process (detection of sabotage) acts as an algorithmic trigger for an exponential information cascade. Because this punitive decay mechanism ($-\kappa R$) strictly negates the artificial connection weights acquired in Phase 1, the short-term payoff maximization is completely nullified.

The robustness validation explicitly demonstrates that the exponential decay fundamentally overwhelms any initial advantage in reputation. Regardless of whether the agent started as a highly trusted authority (Pareto/Lognormal) or an average trusted peer (Uniform), the expected reputation deterministically collapses below θ_R , isolating over 97.7% of sabotaging nodes.

Therefore, empirical calibration of initial reputation is mathematically irrelevant to the ultimate topological outcome. This structural collapse invalidates the connection weights, mathematically destroying the Myopic Nash Equilibrium established in Phase 1 and successfully initiating the preconditions for Topological Death (Hypothesis 3). The network’s self-regulation is proven to be a topologically deterministic response to deceptive signaling.

Supplementary Material: Detailed Computational Results for Phase 3 (Hypothesis 3)

This supplementary document details the integration of the temporal phases and the formal mathematical evaluation of the Nash Equilibrium Phase Transition (Hypothesis 3). Utilizing the Hamilton-Jacobi-Bellman (HJB) utility framework, the computational model calculated the cumulative expected utility (J) over an extended time horizon ($T_{max} = 20.0$) with a continuous discount rate ($r = 0.05$). The objective was to ascertain the absolute rationality of the parasitic sabotage strategy when subjected to the algorithmic immune response of the network topology.

S7. Integration of Expected Utility and the Phase Transition

Table 5 maps the trajectory of the expected instantaneous payoff rate for the sabotage strategy ($\mathbb{E}[\pi_{Sabotage}]$) alongside the cumulative discounted expected utilities for both strategies.

During the initial phase ($t < 5.0$), the sabotage strategy extracts a highly positive instantaneous payoff, driving an initial surge in its cumulative utility (J_s) and establishing a Myopic Nash Equilibrium. However, the topological death triggered in Phase 2 results in a perpetual negative payoff rate ($-c_k$), which becomes evident as $\mathbb{E}[\pi_{Sabotage}]$ collapses into the negative domain.

By $t = 6.72$, the integral of these continuous isolation penalties strictly equals and subsequently overcomes the initial bounded surplus, marking the exact Phase Transition Time (Crossover Point). Evaluated over the full temporal horizon, the infinite sequence of isolation penalties fundamentally negates the transient gains. The terminal evaluation confirms a final cumulative expected utility of 14.8564 for the Honest strategy and 0.6362 for the Sabotage strategy.

S8. Cross-Distributional Robustness of the Utility Differential

To confirm that the phase transition is topologically invariant, the simulation evaluated the utility differential ($\Delta\mathbb{E}[J] = \mathbb{E}[J_{Sabotage}] - \mathbb{E}[J_{Honest}]$) across 1,000,000 independent trials for Uniform, Lognormal, and Pareto initial distributions.

As documented in Table 6, a high initial reputation (e.g., Pareto distribution) slightly delays the topological death and extends the short-term payoff spike. This is reflected in the marginally

Table 5: Temporal Evolution of Instantaneous Payoff and Cumulative Expected Utility

Time (t)	$\mathbb{E}[\pi_{Sabotage}]$	$\mathbb{E}[J_{Honest}]$	$\mathbb{E}[J_{Sabotage}]$
0.00	1.4883	0.0140	0.0149
2.50	1.3878	2.9758	6.2392
5.00	-0.1166	5.3888	7.3251
6.72	-0.5559	6.8695	6.8635
10.00	-0.8809	9.3610	5.2464
15.00	-0.9841	12.4472	2.7116
20.00	-0.9978	14.8564	0.6362

Table 6: Cross-Distributional Evolution of the Utility Differential ($\Delta\mathbb{E}[J]$) up to $T_{max} = 25.0$

Time (t)	Uniform Distribution	Lognormal Distribution	Pareto Distribution
0.00	0.0009	0.0009	0.0009
5.00	1.9372	1.9454	2.0218
6.72	0.0000	0.0125	0.0815
10.00	-4.1272	-4.1043	-4.0305
15.00	-9.7499	-9.7221	-9.6519
20.00	-14.2343	-14.2058	-14.1360
25.00	-17.7371	-17.7085	-17.6388

delayed phase transition times: $t \approx 6.72$ for the Uniform baseline, $t \approx 6.73$ for the Lognormal distribution, and $t \approx 6.79$ for the Pareto distribution.

However, this pre-existing authority cannot prevent the ultimate structural collapse. Evaluated over a long temporal horizon, the continuous topological penalty mathematically guarantees that the utility differential crosses zero. The final utility differentials strictly converge into the negative domain: -17.7371 (Uniform), -17.7085 (Lognormal), and -17.6388 (Pareto).

S9. Comprehensive Discussion of Phase 3 Mechanics

The computational verification rigorously satisfies the condition $\mathbb{E}[J_{Sabotage}] < \mathbb{E}[J_{Honest}]$, indicating that a phase transition to the True Nash Equilibrium is successfully achieved. The long-term inequality strictly converges to $\mathbb{E}[J_{Sabotage}] < \mathbb{E}[J_{Honest}]$, proving that honest cooperation is the sole mathematically rational and Pareto-optimal Nash Equilibrium.

The cross-distributional robustness analysis further confirms that this strict inversion proves that the phase transition to the True Nash Equilibrium is topologically invariant and completely independent of empirical calibration. Highly connected nodes or agents endowed with supreme initial trust can only prolong the Myopic Nash Equilibrium by a fraction of a temporal unit before the immune cascade inevitably triggers systemic isolation. Consequently, parasitic sabotage is mathematically proven to be a topologically self-defeating strategy in scale-free information ecosystems.

Supplementary Material: Detailed Computational Results for the Integrative 4-Phase Dynamics

This final supplementary section details the integrative 4-phase stochastic model, which evaluates the complete temporal lifecycle of the parasitic sabotage strategy. By synthesizing the localized resource extraction (Phase 1), the network immune response (Phase 2), and inter-cluster synchronization (Phase 3), we mathematically define the thermodynamic trajectory toward terminal Topological Death (Phase 4).

S10. Granular Metrics of the 4-Phase Systemic Transition

Table 7 provides the granular temporal evolution of the system state variables utilizing the Uniform distribution as a baseline (1,000,000 independent paths). The metrics track the expected reputation in the superior cluster ($\mathbb{E}[R_{sup}]$), the original cluster ($\mathbb{E}[R_{orig}]$), the instantaneous payoff rate ($\mathbb{E}[\pi_s]$), and the proportion of the ensemble transitioning into subsequent phases.

The computational data explicitly isolates the transient nature of the Myopic Nash Equilibrium. The instantaneous payoff peaks at $\mathbb{E}[\pi_s] = 5.4736$ at $t = 1.00$. However, as $\mathbb{E}[R_{sup}]$ crosses the synchronization threshold ($\theta_{sync} = 0.6$) near $t = 1.38$, the inter-cluster transmission of warning signals triggers Phase 3. This bilateral reputation decay forces the system into Phase 4, crossing the maintenance threshold ($\theta_R = 0.2$) and causing the instantaneous payoff to invert into the negative domain ($\mathbb{E}[\pi_s] = -0.1072$) by $t = 6.00$. The ultimate convergence yields an isolation rate of 99.69% and an asymptotic payoff rate converging to the maintenance penalty ($-c_k \approx -1.0$).

S11. Cross-Distributional Validation of Topological Death

To rigorously exclude the possibility of empirical calibration bias, the phase transition parameters were evaluated across egalitarian (Uniform) and heavily stratified (Pareto) network structures.

Table 7: Temporal Evolution of System State Variables (Uniform Distribution Baseline)

Time (t)	$\mathbb{E}[R_{sup}]$	$\mathbb{E}[R_{orig}]$	Payoff Rate ($\mathbb{E}[\pi_s]$)	Phase 3 Rate (%)	Phase 4 Rate (%)
0.00	0.8998	0.8999	3.7459	0.00	0.00
1.00	0.6893	0.6794	5.4736	29.74	18.38
2.00	0.4731	0.4608	3.4297	52.86	45.22
4.00	0.2232	0.2135	0.9912	78.81	75.40
6.00	0.1098	0.1012	-0.1072	90.50	88.97
10.00	0.0358	0.0278	-0.8188	98.07	97.76
15.00	0.0193	0.0116	-0.9750	99.74	99.69

Table 8 contrasts the decay of the superior cluster reputation and the corresponding Topological Death rates.

Table 8: Cross-Distributional Convergence of the Terminal Phase (Uniform vs. Pareto)

Time (t)	Uniform Baseline		Pareto (High Initial Authority)	
	$\mathbb{E}[R_{sup}]$	Phase 4 Death Rate	$\mathbb{E}[R_{sup}]$	Phase 4 Death Rate
0.00	0.8998	0.00%	0.9750	0.00%
2.00	0.4731	45.22%	0.5032	44.82%
6.00	0.1098	88.97%	0.1149	88.80%
15.00	0.0193	99.69%	0.0194	99.70%

The Pareto distribution assigns an extreme initial structural authority ($\mathbb{E}[R_{sup}] \approx 0.9750$). However, the mathematical mechanics of the Poisson-triggered information cascade scale proportionally with the target’s reputation vector. Consequently, the exponential decay rapidly consumes this structural buffer. As illustrated in Table 8, the differences in reputation and isolation rates become statistically negligible by $t = 2.00$. At the terminal evaluation ($T_{max} = 15.0$), both cohorts exhibit a complete structural collapse, resulting in statistically identical Phase 4 Death Rates (99.69% and 99.70%).

S12. Concluding Summary of the Supplementary Validation

The granular data presented in these supplementary materials (S1 through S11) provides exhaustive computational proof for the central thesis of the study.

The evaluation of Phase 1 mathematically confirmed that deceptive signaling (parasitic sabotage) is locally rational under incomplete information, strictly satisfying the conditions for a Myopic Nash Equilibrium by generating a massive, localized surge in expected payoff. However, the subsequent analysis of Phase 2 and Phase 3 demonstrated that this localized advantage functions entirely as an algorithmic vulnerability within the host network.

The integration of the Poisson detection process and inter-cluster synchronization mechanisms proved that the network topology possesses an autonomous, thermodynamic self-regulation capacity. Once the reputation cascade breaches the critical boundary $\theta_{sync} = 0.6$, the resulting structural isolation is mathematically deterministic and topologically invariant. Neither initial network cen-

trality nor supreme pre-existing trust (e.g., Pareto distribution) can halt the exponential decay induced by the immune cascade.

Ultimately, the Hamilton-Jacobi-Bellman utility integration executed in Hypothesis 3 decisively resolves the game-theoretic dilemma. The continuous topological maintenance penalties exacted during complete isolation systematically negate all prior transient gains, establishing an irreversible phase transition where the cumulative utility differential strictly crosses into the negative domain ($\Delta\mathbb{E}[J] \ll 0$). Therefore, the supplementary data categorically validates that continuous honest cooperation is not a merely ethical construct, but the sole mathematically rigorous and Pareto-optimal True Nash Equilibrium within scale-free information ecosystems.